

EXP-007 — Pretrained Semantic Retrieval

Testing whether a genuinely pretrained embedding model bridges the vocabulary gap that BM25 cannot.

EXECUTIVE RESULT

The vocabulary-mismatch hypothesis was not supported. Standalone dense retrieval is worse than BM25 (0.425 vs 0.475) and far worse at depth — 5 spans unreachable at 300 against BM25's 1. **Neither of its two rescues was a vocabulary rescue:** both already had ≥92% of their query terms present. **AN-003, the canonical case, was not rescued.**

The unexpected finding is complementarity. **BM25 + dense RRF reaches 0.600 and 11/20** — the best configuration measured in this project — because the two retrievers fail on different questions.

0.920

Mean lexical overlap of the spans dense "rescued" — the query terms were already there, so this was a ranking fix, not a vocabulary fix

AN-003: —

Still absent at depth 300 under dense, though dense ranked its *document* first

Model, selected before any result was seen

identifier	fasttext-wiki-news-subwords-300 (gensim-data)
origin / revision	facebookresearch/fastText wiki-news-300d-1M-subword gensim-data release asset, sha256 be48d40d6c67dbebe4b3eea22ab7dd7c7efbdb977d88...
training corpus	Wikipedia 2017 + UMBC webbase + statmt.org news (~16B tokens)
dimension / metric	300, cosine; pooling: mean of L2-normalized in-vocabulary token vectors
corpus-fitted?	No — trained entirely independently of this project

Why not a transformer encoder: measured, not assumed — `huggingface.co` returns `CONNECT tunnel failed, 403`, and every embedding API is equally blocked. GitHub release assets are reachable, which is how a pretrained model was obtained at all. Corpus-fitted TF-IDF+SVD was forbidden and was not used.

INSTRUMENT STRENGTH — THIS QUALIFIES THE NEGATIVE RESULT

This is a genuinely pretrained model but a **static word-embedding** one. Mean-pooled static vectors are order-insensitive and wash out over long chunks. A positive result would have been strong evidence; the negative result obtained is **weak** evidence against the hypothesis. It shows this class of embedding cannot bridge the gap — not that a transformer could not. The hypothesis is **unsupported, not falsified**.

Results

Configuration	macro	full	spans@10	doc	MRR	a@10	a@20	a@50	a@100	a@300	ms
A — BM25 control	0.475	9/20	10/22	0.825	0.280	12	8	3	2	1	381
B — pretrained dense	0.425	8/20	9/22	0.725	0.360	13	9	9	7	5	62
C — BM25 + dense RRF	0.600	11/20	13/22	0.825	0.326	9	8	5	2	2	413

Dense has the **highest MRR** despite the lowest macro recall — when it finds evidence it ranks it near the top — but the **worst deep reachability**. EXP-007A reproduced the frozen BM25 baseline exactly (0.475, 9/20, identical hit ordering), so the comparison is sound.

BM25 vs dense — paired

Quadrant	n	cases
BM25 correct / dense correct	6	AN-009, OA-001, OA-002, OA-003, OA-005, OA-008
BM25 correct / dense wrong	3	AN-005, OA-004, OA-007
BM25 wrong / dense correct	2	AN-002, AN-007
both wrong	9	AN-001, AN-003, AN-004, AN-006, AN-008, AN-010, AN-011, AN-012, OA-006

Comparison	Δ macro	rescued	regressed	net
BM25 -> dense	-0.050	AN-002, AN-007	AN-005, OA-004, OA-007	-1
BM25 -> RRF	+0.125	AN-002, AN-004, AN-007	OA-004	+2
dense -> RRF	+0.175	AN-004, AN-005, OA-007	—	+3

What dense actually bought

No dense rescue was a vocabulary-mismatch rescue. Both rescued spans already had >=92% of their query terms present in the chunk; dense corrected a BM25 ranking failure, not a lexical coverage failure. The one genuine vocabulary-mismatch case, AN-003, was not rescued.

Rescue	BM25 → dense	lexical overlap	mechanism
AN-002	27 → 1	12/13	topical coherence, not a vocabulary gap
AN-007	18 → 2	11/12	topical coherence, not a vocabulary gap

Both live in the same 3,327-character `HTTP errors` chunk — a homogeneous list of status codes. BM25 ranked it 27th and 18th because its length normalization penalises a long chunk; dense matched the chunk's topic to the query's. **Dense fixed a BM25 ranking failure, not a lexical coverage failure.**

Per-question ranks

Case	Category	BM25	dense	RRF
AN-001	exact_lookup	19	242	38
AN-002	exact_lookup	27	1	2
AN-003	exact_lookup	—	—	—
AN-004	exact_lookup	12	17	4
AN-005	exact_lookup	4	—	8
AN-006	exact_lookup	29	63	56
AN-007	exact_lookup	18	2	4
AN-008	normal	12	123	25
AN-009	version_conflict	1	1	1
AN-010	version_conflict	49	—	97
AN-011	ambiguous	54	14	29
AN-012	multi_hop	47,117	1,—	2,—
OA-001	exact_lookup	4	1	2
OA-002	exact_lookup	2	1	2
OA-003	exact_lookup	6	1	3
OA-004	exact_lookup	5	73	17
OA-005	exact_lookup	3	1	1
OA-006	multi_hop	1,29	18,—	2,55
OA-007	normal	1	11	2
OA-008	exact_lookup	1	10	1

AN-003 — the canonical case

Retriever	evidence rank	doc rank	@10	@20	@50	@100	@300
A — BM25 control	—	6	no	no	no	no	no
B — pretrained dense	—	2	no	no	no	no	no
C — BM25 + dense RRF	—	2	no	no	no	no	no

Outcome: failure. But the trace shows why, and it is not what the hypothesis predicted. Dense put the correct *document* at rank 1 (BM25: 6) and its top five hits were all `Create a Message Batch` chunks at cosine **0.9417** — the

model *did* associate the query with the right material. It failed to surface the answer because that chunk is 3,449 characters of heterogeneous parameter documentation and its mean-pooled vector is diluted.

The evidence never entered a rerankable range under any configuration, so **a reranker could not rescue AN-003**.

New observation: chunk length predicts dense reachability (median 897 chars reachable vs 1,754 unreachable) in a way it did not predict BM25 reachability. EXP-005 tested chunk size against BM25 alone and found nothing; it has never been tested against dense retrieval.

Fusion

Preregistered before any result: pool 50 per retriever, `rrf_k=60` , `top_k=10` . Not tuned; no sweep run. RRF beats both parents and restores what each broke — AN-005 (dense lost it entirely) returns at rank 8, OA-007 at rank 2.

The fusion regression the brief warned about did occur: OA-004 sits at BM25 rank 5, dense rank 73, and **RRF rank 17** — dragged outside `top_k` by a weak dense rank. One case, reported rather than absorbed into the average.

EXP-NULL — still blocked

Retried. Still no generation credential and the provider host is egress-blocked; the results file records `status: "blocked"` with the exact error. Retrieval remains uncalibrated against what the model already knows.

Updated root-cause conclusion

#	Hypothesis	Verdict
1	Oversized chunks hide evidence	falsified — 0 rescued
2	Missing structural context	falsified — Δ0.000
3	Lexical vocabulary mismatch	unsupported — no rescue was a vocabulary rescue; canonical case failed

What EXP-007 established instead: **BM25 and dense fail on disjoint questions**. The residual failures look less like a vocabulary problem and more like a **retrieval-unit** problem — AN-003's evidence is a 57-character sentence inside a 3,449-character heterogeneous chunk, and both retrievers fail on it for their own reasons.

Limitations

- The instrument is weak** — static vectors, not a transformer encoder, because every transformer host is blocked. The negative result is weak evidence.
- n = 20**. One case is five percentage points of macro recall. Every movement here is 1–3 cases. Nothing supports "dense beats BM25" or the reverse in general.
- Mean pooling is crude** and order-insensitive; some failures are the pooling, not the vectors.
- 117 chunks (0.8%) have all-zero embeddings** and are unreachable by dense.
- EXP-NULL never ran**.
- No parameter was swept**; BM25 and RRF settings are fixed.

What is justified next

1. **Bounded chunking × dense retrieval.** The strongest lead: EXP-005 already built the bounded chunk set, and EXP-007 shows dense reachability tracks chunk size. A cheap 2×2 that directly targets AN-003.
2. **A transformer retrieval encoder, if egress ever permits.** The hypothesis is unsupported, not falsified, and the instrument is the reason.
3. **Not a reranker.** AN-003 is absent at depth 300 under every configuration including fusion. The condition that would justify one — evidence routinely at rank 15–100 but not top 10 — is not what the data shows.

Promotion decision: the frozen baseline does not change (control chunking, no enrichment, BM25, top_k=10).

RRF is the best-measured configuration and the leading candidate, but it is a +2-case movement on a 20-case development set, it carries a real regression, and it doubles latency. Not enough to change a production baseline.

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